

Full Stack Development with MERN

1. Introduction

Project Title: ShopEZ – E-Commerce Web Application

Team Members:

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2. Project Overview

ShopEZ is a full-stack e-commerce web application that enables users to browse products, register, log in, manage carts, and place orders. Administrators can manage products, users, and orders.

Features

- User Registration & Login
 - Product Listing
 - Product Search
 - Shopping Cart
 - User Profile
 - Admin Dashboard
 - Product Management
 - User Management
 - Order Management
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3. Architecture

Frontend

- React.js
- HTML
- CSS
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB Atlas / MongoDB
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4. Setup Instructions

Prerequisites

- Node.js
- MongoDB
- Git

Installation

```
git clone <repository-link>
cd ShopEZ
npm install
npm run dev
```

5. Folder Structure

```
ShopEZ
|
├── client
|   ├── src
|   ├── components
|   ├── pages
├── server
|   ├── controllers
|   ├── models
|   ├── routes
└── package.json
```

6. Running the Application

Frontend:

npm run dev

Backend:

npm start

Application URL:

http://localhost:5173

7. API Documentation

Method Endpoint Description

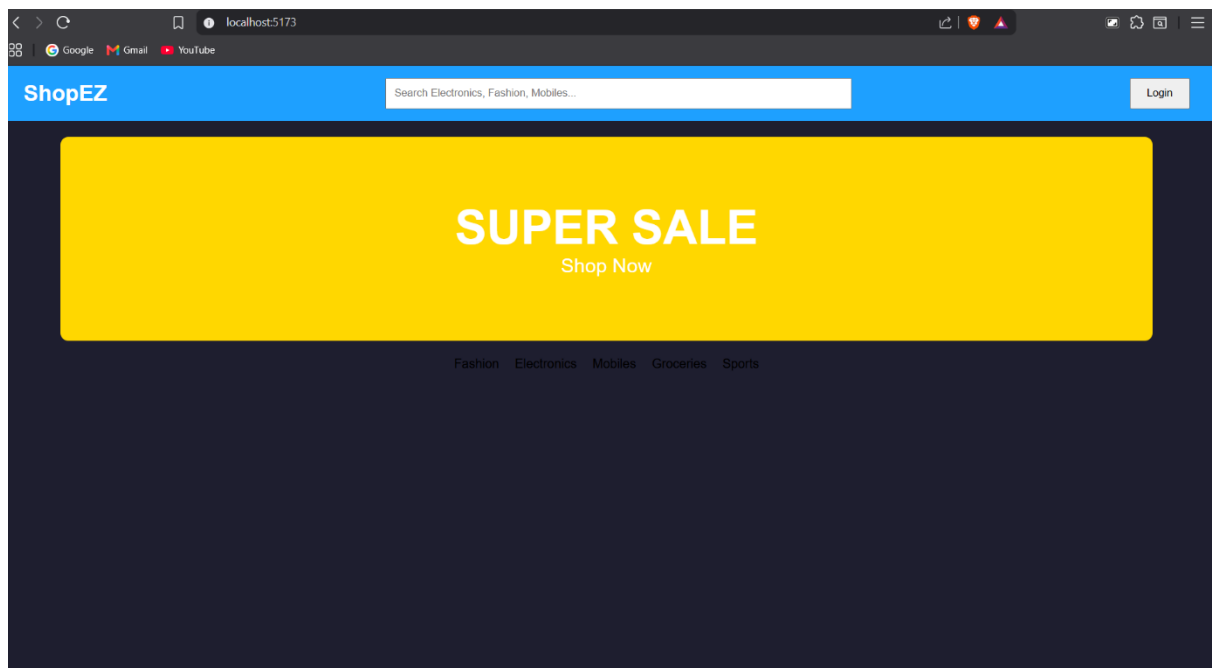
POST	/register	User Registration
POST	/login	User Login
GET	/products	Fetch Products
GET	/cart	View Cart
GET	/profile	User Profile
GET	/admin	Admin Dashboard

8. Authentication

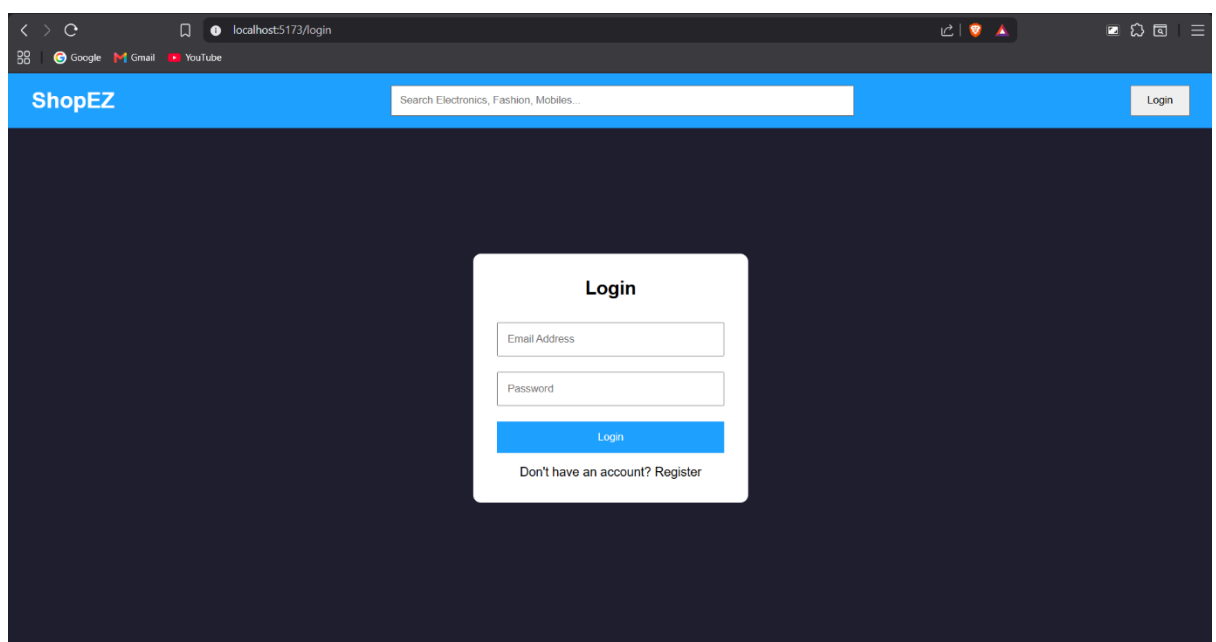
- Secure Login System
 - Session/JWT Authentication
 - Protected Admin Routes
 - Role-Based Access Control
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9. User Interface

- Home Page



- Login Page



- Registration Page

localhost:5173/register

ShopEZ

Search Electronics, Fashion, Mobiles...

Login

Register

Username

Email Address

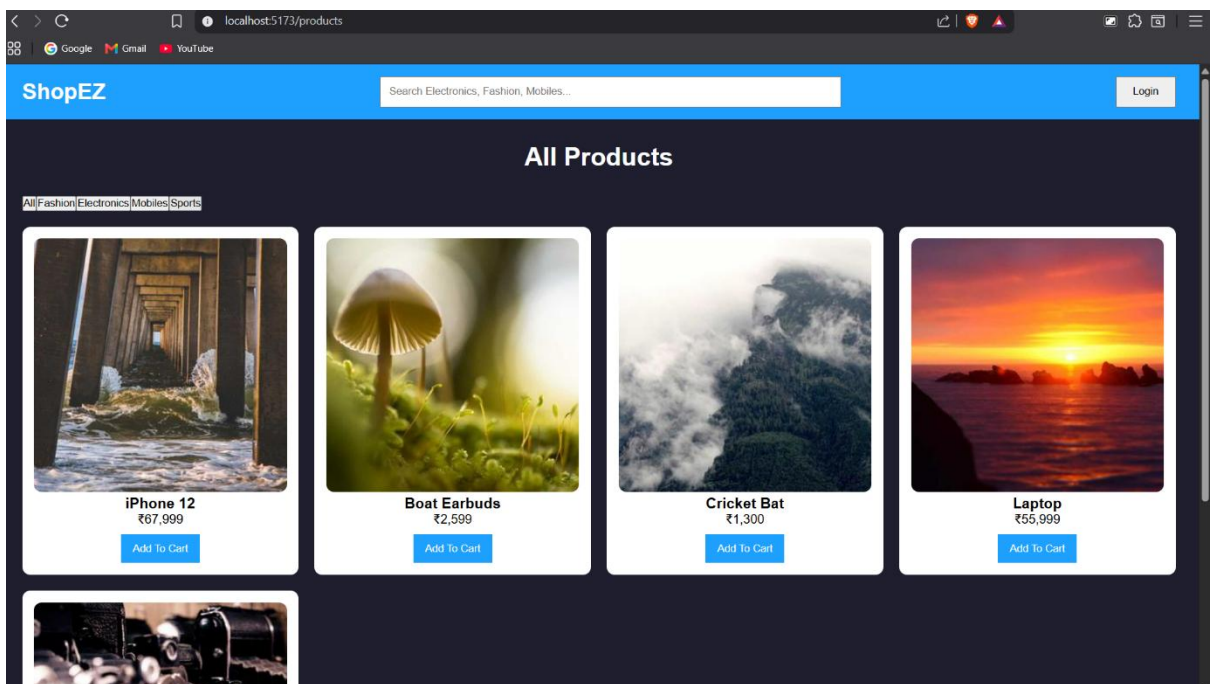
Password

User Type ▾

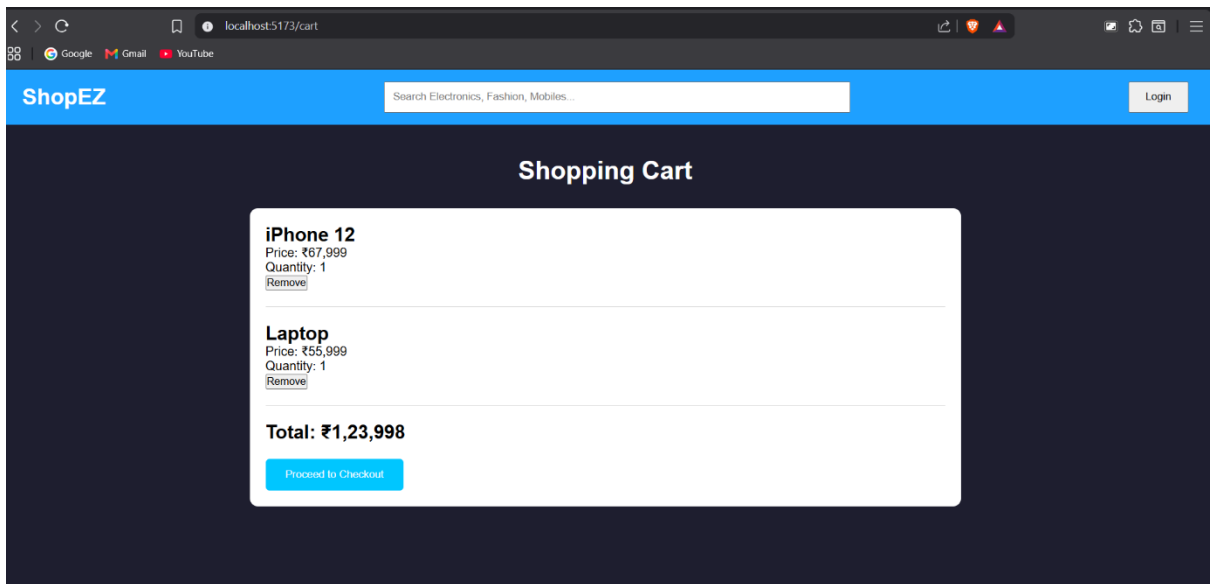
Register

Already registered? Login

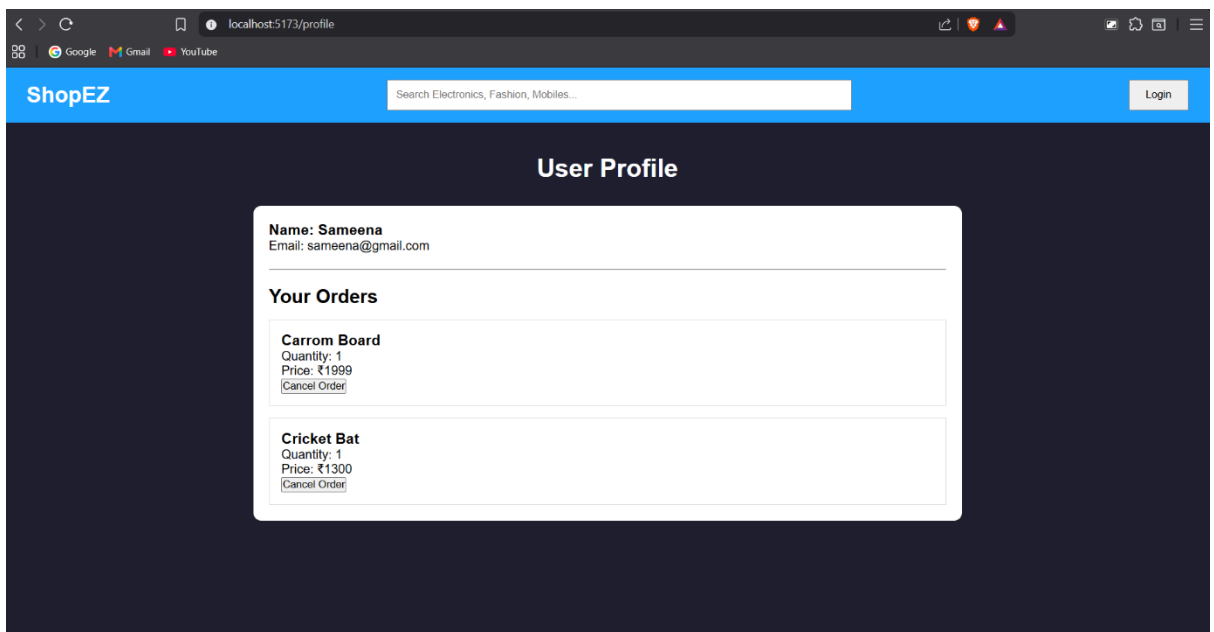
-
- Products Page



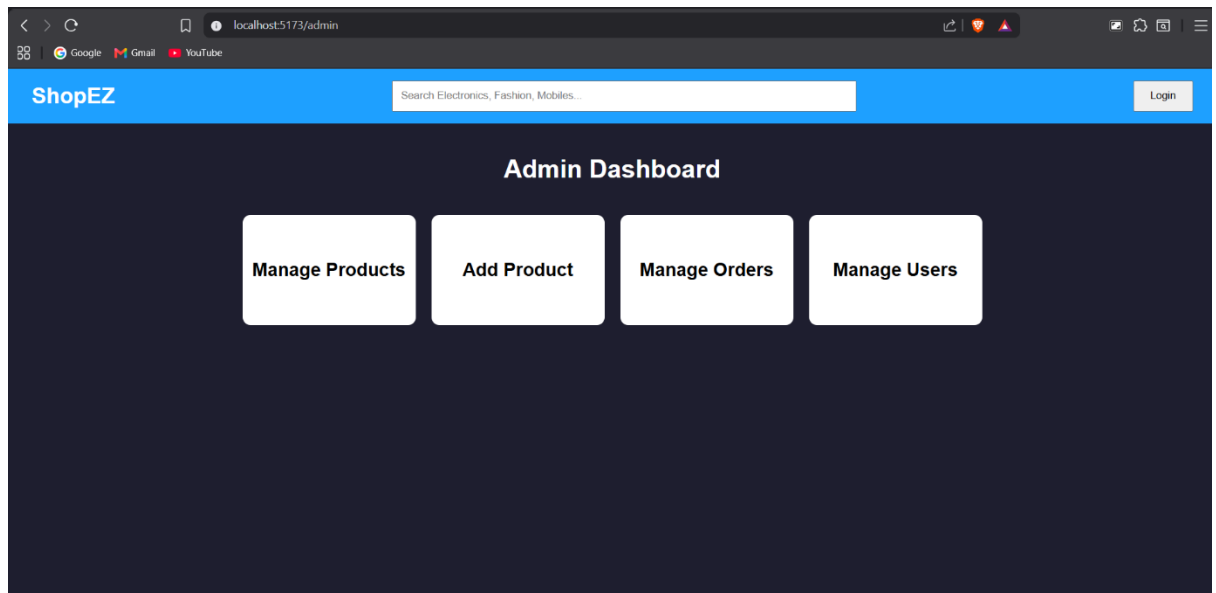
-
- Cart Page



- Profile Page



- Admin Dashboard



10. Testing

Testing Performed

- Registration Testing
- Login Testing
- Product Display Testing
- Cart Testing
- Admin Module Testing

Result

All major functionalities tested successfully.

11. link

GitHub Repository

<https://github.com/sameenatanwaz/VIP-C2-Sameena-Tanwaz>

12. Known Issues

- No major issues identified during testing.

13. Future Enhancements

- Online Payment Gateway
- Wishlist Feature
- Product Reviews & Ratings
- Order Tracking
- Email Notifications
- Mobile Responsive Enhancements